

Éamonn Needham

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Education

Arizona State University

Graduated: May 2025

Ph.D. in Geological and Planetary Sciences

Dissertation:

A Multiscale Investigation of Geochemical and Physical Heterogeneities in Igneous Processes

Advisors: Mélanie Barboni, Amanda Clarke

University of Oregon

Graduated: June 2019

Bachelor of Science, Phi Beta Kappa, Sigma Gamma Epsilon

Honors Thesis: *Experimental Study of Microlite Crystallization During Magma Ascent*

Current Research Focuses

- Experimental investigations into progressive quartz cohesion-induced tremor
- Re-assessing Ti-in-Quartz geothermometry and geobarometry methods
- Ochellar quartz as a window into magma plumbing systems
- Ca-isotope diffusion in plutonic systems
- Conditions of magma storage in the upper crust.
- Formation and chemistry of early lunar magmas.
- Trace element geochemistry of zircon, plagioclase, and volcanic glass.
- Developing new analytical tools to source archeologically significant cryptotephra.
- Upper conduit fragmentation mechanisms and particle formation from basalt to rhyolite.

Papers

Peer Reviewed Publications:

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| 2025 | Barboni, M., Needham, É. , Bell E., Chen, H., Trail D., <i>Pervasive impact modification of pristine lunar clasts</i> , Nature Communications. |
| 2022 | Yuan, Q., Zhang, C., Cheng, F., Cao, X., Needham, É. , Zheng, H., Lü, X., <i>In-situ U–Pb dating of zircon coronas, Sr–Nd–Hf isotopes and petrological constraints of the Daxigou anorthosite complex, NW China</i> , Gondwana Research. |

Papers in Press:

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| July 2025 | Needham, É. , Clarke, A., Roggensack, K., Zawacki, E., <i>Fragmentation and scoria agglutination formation mechanics in a highly explosive basaltic eruption</i> , JVGR. |
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Papers in Prep:

- In Prep **Needham, É.**, Barboni, M., *Trace Element Diffusion in Plagioclase coupled to zircon thermometry document simultaneous Warm and Cold Magma Storage in Arc Volcanoes*, EPSL.
- In Prep Hirniak, J., **Needham, É.**, Karkanas P., Barboni, M., Smith, E., Campisano C., Marean C., *Archeological Stratigraphic Provenance of Crypto tephra from a Super volcanic eruption*, Journal of Quaternary Science.
- In Prep Hirniak, J., **Needham, É.**, Karkanas P., Smith, E., Campisano C., Marean C., *The effects of archaeological fires on volcanic glass*, TBD.

Presentations

Oral:

- September 2026 **Needham, É.**, Barboni, M., *Discovering Lost Lunar Magmas*, Sacramento State University Seminar, Sacramento, CA.
- August 2026 **Needham, É.**, *Rediscovering Lost Lunar Magmas*, Astronomy on Tap, Oakland, CA.
- March 2026 **Needham, É.**, Zawacki, E., Clarke, A., Roggensack, K., *Heterogeneous Subsurface Fragmentation Regimes in Highly Explosive Basaltic Eruptions*, UC Davis Lunch Talk, Davis, CA.
- May 2026 **Needham, É.**, Barboni, M., Trail, D., Bell, E., *How Pristine Are Lunar Samples? Rethinking the Sample Record of the Early Moon*. UC Davis Department Seminar, Davis, CA.
- May 2025 **Needham, É.**, *A Multiscale Investigation of Geochemical and Physical Heterogeneities in Igneous Processes*, Dissertation Defense, ASU, Phoenix, AZ.
- March 2024 **Needham, É.**, Barboni, M., Trail, D., Bell, E., *Probing the Cogeneity of Minerals and Glasses in Lunar Breccias Using Aluminum in Zircon and Glass*, LPSC, Houston, TX.
- March 2024 Hirniak, J., Smith, E., Johnsen, R., **Needham, É.**, Barboni, M., Marean C., *The potential of crypto-tephrochronology studies at paleoanthropological sites throughout Africa*, Paleoanthropology Society, Los Angeles, CA.
- October 2023 Hirniak, J., **Needham, É.**, Karkanas P., Smith, E., Campisano C., Marean C., *The effects of archaeological fires on volcanic glass shards: An experimental approach*, Sci-X, Reno, NV.
- July 2023 **Needham, É.**, Barboni, M., *Sr and Mg diffusion models of plagioclase show potential long melt residence time and complex mush structure beneath arc volcanoes*, Goldschmidt, Lyon, France.

Posters:

- October 2026 **Needham, É.**, Watkins, J., *Titanium-in-Quartz: Time series experiments reveal a possible non-classical crystallization pathway*, ExPet 2026, Washington D.C.
- July 2026 **Needham, É.**, Watkins, J., *Titanium-in-Quartz: Time series experiments reveal a possible non-classical crystallization pathway*, Goldschmidt, Montreal, Canada.
- April 2026 **Needham, É.**, Watkins, J., *Beyond Equilibrium: Amorphous Precursor-Mediated Ti Uptake and Its Implications for TitaniQ Thermobarometry*, UC Davis Postdoc Symposium, Davis, CA.
- September 2022 **Needham, É.**, Zawacki, E, Clarke, A, Roggensack, K, *Micro-Textural Analysis of the Highly Explosive Basaltic Eruption at Tecolote, Pinacates Volcanic Field, Mexico*, Chapman Conference on Distributed Volcanism, Flagstaff, AZ
- December 2020 **Needham, É.**, Marquardt, M, Barboni, M, *New perspectives on the use of triple quadrupole ICP-MS (laser and solution) systems for geochemical studies*, American Geophysical Union, Online.
- December 2019 Trafton, K, Giachetti, T, **Needham, É.**, *Making Pumice: a Post-Fragmentation Process?*, American Geophysical Union, San Francisco, CA.
- May 2019 **Needham, É.**, Giachetti, T., Watkins, J., *Experimental Study of Microlite Crystallization at Mono Craters*, Volc-OR, Eugene, OR.
- December 2018 **Needham É.**, Watkins J., Garcia V., Gardner J., Giachetti T., *Experimental study of microlite crystallization during magma ascent*, American Geophysical Union, Washington D.C.
- December 2018 Olsen E, **Needham É.**, Watkins J, Devriendt L, *The effect of salinity on the activity of carbonic anhydrase and $\delta^{18}\text{O}$ of carbonates*, American Geophysical Union, Washington D.C.
- November 2018 Wiejaczka J., Giachetti T., Watkins J., Gardner J, Trafton K, **Needham É.**, *Obsidian pyroclasts: pre-eruptive vanguard magma, or juvenile clasts formed after fragmentation?*, Geological Society of America, Indianapolis, IN.
- August 2018 Watkins K., Olsen E., **Needham É.**, *Calcite precipitation from saline solutions: How dissolved NaCl affect DIC speciation, DIC isotopic equilibration, and $\delta^{18}\text{O}$ in calcite*, Goldschmidt, Boston, MA.
- April 2018 **Needham, É.**, Giachetti, T., Watkins, J., *Determining the Mono Craters decompression rate: microlite and bubble number densities*, Volc-OR, Corvallis, OR.
- October 2017 Olsen, E., **Needham, É.**, Watkins, J., *Quantifying the salinity effect on isotope fractionation during inorganic calcite precipitation*, Geologic

Society of America, Seattle, WA.

Undergraduate Research Experience

- Summer 2018 **Field Assistant**, *Physical Volcanology lab*, UO. Supervisor: Thomas Giachetti, Ph.D. Pit digging, pyroclast sample collection, and sample characterization at Newberry Caldera and Crater Lake.
- 2018 – 2019 **Experimental Volcanology Research Assistant**, UO. Supervisors: Thomas Giachetti, Ph.D. and James Watkins, Ph.D. Ran cold-seal-pressure-vessel decompression experiments to reproduce microlite textures in synthetic obsidian.
- 2017 – 2018 **Experimental Volcanology Research Assistant**, UO. Supervisor: James Watkins, Ph.D. Created volatile maps of obsidian pyroclasts from Mono Craters using an FTIR to better understand obsidian pyroclast formation.
- Summer 2017 **Field Assistant**, *Earthquake Seismology lab*, UO. Supervisor: Amanda Thomas, Ph.D. Installed geophones on and around Mt. Hood, Oregon.
- 2016 – 2017 **Presidential Undergraduate Research Fellow**, UO. Supervisor: James Watkins, Ph.D. Grew calcite crystals at varying salinities to better understand the role of kinetic effects on the isotopic composition of calcite.
- 2016 – 2017 **Experimental Igneous Petrology Research Assistant**, UO. Supervisor: James Watkins, Ph.D. Designed and ran experiments that explored how liquids un-mix across a thermal gradient using a piston cylinder apparatus.

Instrument Experience

Analytical Instruments:

- 2025-2026 FIB, TEM, Profilometry
- 2019– 2024 **Laser Ablation Triple Quadrupole Inductively Coupled Plasma Mass Spectrometer (ASU)** Lab manager role. Responsible for running the lab, ordering gas, maintenance and repair of the instrument, method development for trace element analysis in plagioclase and glass. Developed an analytical method that allowed for a 9x reduction in ablated volume while maintaining commensurate uncertainties. Enabled the transition to recharge lab status for the archeological cryptotephra community based off the quality of the analytical method developed.
- 2023 – 2024 **Electron Backscatter Diffraction (ASU)**, Analyzed the crystallographic structure of Lunar zircons in difficult Apollo samples, assessed the degree of misorientation to provide spatial context to collected geochemical data.
- 2023 **Secondary Ionization Mass Spectrometer (UCLA)**, ims-1290 Hyperion ion

source. U-Th disequilibria dating and Trace element/REE geochemistry of zircon rims on indium mounts. In-situ thin section Lunar Zircon Pb/Pb dating and Al-in-zircon + Trace element geochemistry.

- 2020-2024 **Electron Microprobe (ASU)**, EDS and WDS of major elements in plagioclase, BSE imaging of scoria micro-textures, X-ray mapping of lunar thin sections to identify zircon, major elements in lunar glasses, and EBSD.
- 2019-2024 **Petrographic Microscope (ASU)**, Characterization of ~45 dacitic thin sections for their componentry, plagioclase textural features, bordering phases, and choosing a representative sampling of >100 plagioclase for further chemical analysis. Picking of zircon separates for SIMS analysis.
- 2019 **Raman Spectrometer (ASU)**, Used to identify high pressure silica polymorphs in lunar Apollo samples.
- 2017 – 2019 **Fourier Infrared Spectrometer (UO)**, Analysis of volatiles in obsidian, installation of an ATR crystal option for transmission spectroscopy and subsequent method development, training of three graduate students on the ATR-FTIR.

Experimental Instruments:

- 2016-2019 **Cold Seal Pressure Vessel Apparatus (UO)**, Experimental design, sample construction and standardization of components, organized manufacture of parts for improved efficiency, running of the apparatus (20+ runs) for my own samples and those of other professors and graduate students. Ran rapid quench, decompression, and stalling experiments to assess the conditions of microlite and bubble growth in synthetic rhyolitic glass.
- 2016-2018 **Experimental Calcite Controlled Growth Apparatus (UO)**, Ran 20+ experiments to grow synthetic calcite under controlled chemical conditions. Experiment planning, method development, clean room work, experiment monitoring and sample analysis.
- 2015-2016 **Piston Cylinder Apparatus (UO)**, Method Development, sample capsule design and construction, troubleshooting, experimental durations for up to two weeks, and training of others on sample construction.

Teaching Experience

- Fall 2026 Collab on Environmental Geochemistry
- Spring 2025 **GLG 103 Lab Teaching Assistant**, Taught three weekly labs with 95 total students. Built lectures for every class to introduce the lab material of the week and worked to modernize the lab material with Google Earth.
- Summer 2022 **Geochronology for High School Teachers**, participated in a workshop for 8 visiting high school teachers to integrate geoscience research into the Next Gen Science standards curriculums for their schools. At this workshop, I demonstrated in-situ laser ablation methods, zircon

geochronology methods, and how to texturally characterize lunar zircons on a petrographic microscope.

- Spring 2021 **GLG 103 Lab Teaching Assistant**, Taught two weekly labs with 64 students. Built lectures for every class to introduce the lab material of the week. For the two GLG 103 labs, my student evaluations rating was 4.8 and 4.3 out of 5.
- Spring 2021 **GLG 101 Guest Lecturer**, During the term, I developed and taught an hour-long lecture to 100+ students, titled “Volcanoes-Chapter Six”, incorporating lessons from the textbook, real life examples, and my own volcanology research.
- Spring 2021 **GLG 101 Teaching Assistant**, Responsible for emails and grade inquiries of >100 students, as well as running weekly in-class exercises. For this class, my course evaluation grade was 4.8 out of 5.
- Winter 2018 **Virginia Homeschool Collective**, was invited to give a series of geology lectures to ~ 10 students between the ages of 7 and 15 years old in a homeschool collective in rural Virginia. Topics included mineralogy, local geology, and formation of the Appalachian Mountains, and culminated in a guided field trip to hunt for fossil shark teeth in Potomac River deposits.
- Fall 2017 **Volunteer Lecturer at Dos Rios Elementary, Oregon** over the course of several weeks, I built and presented lectures on mineralogy, volcanoes in our backyard, and earthquake hazard at three elementary science classes, as well as doing demonstrations of clay box topography maps, pancake subduction, and pizza box mineralogy collections, that culminated in guiding a field trip to a local outcrop to teach the basics of field geology.

Miscellaneous Skills:

2022-2024	Mineral Separation/ Sieving/ Panning/ Picking
2021-2022	Rock Crushing
2020-2026	Python Language Coding
2017 – 2024	Public Speaking
2017	Total Station Land Surveying
2017-2023	Physical volcanology eruption characterization
2016 – 2019	Arc Welding
2016 – 2019	Lathe work
2016 - 2023	Carbon Coating
2016 – 2024	Epoxy Puck creation and polishing

Academic Honors

- 2024 Nininger student travel award
- 2023 SESE Summer Exploration Grant
- 2022 GPSA Travel Grant
- 2021 ARC Honorable Mention
- 2020 SIMS UCLA Workshop Award Recipient
- 2018 CURE Undergraduate Conference Travel Award

- 2018 Geo-Emeritus Professor Award
- 2018 Barry Goldwater Fellowship Honorable Mention
- 2018 UROP Mini-Grant
- 2018 National Association of Geoscience Teachers Fellowship
- 2018 Stovall Memorial Award for meritorious performance and promise
- 2017 Geology Emeritus Field Camp Award
- 2017 Walter Youngquist Fellowship Award
- 2016-2017 Presidential Undergraduate Research Scholarship

Service and Outreach

- Volcanoes and Llamas-Puppetry in the Classroom
- Land Survey of Gold Mountain for future Geologic Interpretive Site
- Invited Panelist for current Presidential Undergraduate Research fellows.
- UO Geology Department graduation invited speaker.
- Sigma Gamma Epsilon (Geologic Honor Society) **President** (2018-2019).
- Sigma Gamma Epsilon **faculty liaison**, charter member (2017-2018).
- UO Geology Club **President** (2017-2018).
- Dos Rios Elementary School Breakfast Buddies
- Dos Rios Elementary School volunteer lecturer
- Wallowa Mountains Field Camp Public Forum Presentation
- Science Factory Halloween Mad Scientist volunteer
- Volc-OR Conference **Undergraduate Liaison**
- UO department mineral and meteorite tour guide
- UO Gem and Mineral Showcase organizer